

Luke Melendy

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Education

University of New Hampshire

Durham, NH

Bachelor of Science in Computer Science

Expected Graduation: May 2028

- **Relevant Coursework:** Data Structures, Systems Programming, Operating Systems, Computer Architecture, Software Engineering, Computer Networks

Technical Skills

Languages: Java, Python, C, Kotlin, TCL, HTML/CSS

Tools: Git, Bitbucket, Jira, Linux, Vim, VS Code, Wireshark, AI Workflow tools

Technical Areas: Test Automation, Network Protocol Testing, Backend Development, Debugging, Agile Development

Protocols/Concepts: IPv6, DHCPv6, IPsec, Data Structures, Operating Systems

Experience

Software Developer

May 2025 – Present

UNH InterOperability Laboratory

Durham, NH

- Serve as a primary Java contributor on a 4-person team developing and maintaining IOL-INTACT, an internal network protocol testing platform.
- Develop Java-based testing infrastructure supporting protocol conformance, interoperability workflows, and long-term maintainability.
- Write and debug IPv6-related network protocol test scripts using Kotlin and TCL, including DHCPv6 and IPsec.
- Analyze packet-level behavior using internal packet capture tooling and Wireshark to investigate failures and validate protocol behavior.

Front-End Supervisor

June 2022 – January 2026

Ocean State Job Lot

Hooksett, NH

- Supervise front-end store operations, coordinating associates in a fast-paced customer-facing environment.
- Train team members, resolve customer issues, and maintain reliable daily workflows.

Expertise

AI Agentic Workflows | *Langchain w/python, n8n, Local and Cloud LLM Models*

- Worked with AI research specialists to develop an effective Network protocol testing tools that uses AI to match packet captures against test specs and RFC's
- Continued AI workflow research into lessening token usage as well as the problems that come with using AI as a judge

Network Protocol Test Tool Development | *Java, Javascript, DHCPv6, IPv6, IPsec, Device Interoperability*

- Work on developing features and maintaining the IOL-INTACT test tool - a large scale java project used for validating a devices network protocol functionality to receive IPv6 Ready logo and other relevant certifications

Leadership Activities

President, Esports Program

August 2025 – Present

University of New Hampshire

Durham, NH

- Led the growth of the school esports program by organizing teams, coordinating members, and supporting competitive operations.
- Contributed to securing an official sponsorship and launched multiple new esports teams to expand student participation.

Eagle Scout

June 2022

Boy Scouts of America

Auburn, NH

- Planned and led a community service project involving fundraising, volunteer coordination, budgeting, and project execution.